ModelIt!

Teacher's Guide

How Do Fossils Form?

Clues from Ancient Life

Quick Reference

Grade Level	Grade 3
NGSS Standard	3-LS4-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 3-LS4-1

Analyze and interpret data from fossils to provide evidence of the organisms and the environments in which they lived long ago.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze data from fossil models to provide evidence of how burial conditions affect the fossilization process and what fossils reveal about past environments.
Disciplinary Core Idea	**LS4.A Evidence of Common Ancestry and Diversity** — Some kinds of plants and animals that once lived on Earth are no longer found anywhere. Fossils provide evidence about the types of organisms that lived long ago and the nature of their environments.
Crosscutting Concept	**Scale, Proportion, and Quantity** — Students consider how the enormous scales of time (millions of years) and depth (deep underground) are necessary for the fossilization process to occur.

CAST Testing Alignment

MULTIPLE CHOICE:

A scientist finds a fish fossil in a rock layer high up on a mountain. What does this fossil tell us about that mountain long ago?

CONSTRUCTED RESPONSE:

Explain the steps of how a dinosaur bone that is 65 million years old turned into a fossil we can find today. Use the words 'sediment,' 'pressure,' and 'minerals' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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> **For Your Confidence**
>
> This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Fossils are like nature's time capsules — windows into a world that existed millions or even billions of years ago. Every fossil tells a story about an ancient organism and the environment where it lived.

How Fossils Form

Fossil formation begins when an organism dies and is quickly buried by sediment — sand, mud, or volcanic ash. Quick burial is essential because it protects the remains from scavengers, weather, and decomposition. Over millions of years, more sediment piles on top, creating enormous pressure that compacts the lower layers into sedimentary rock. Meanwhile, mineral-rich groundwater seeps through the remains, slowly replacing the original bone, shell, or wood material with minerals like calcite, silica, and pyrite. The result is a rock replica of the original organism.

Types of Fossils

There are several types of fossils. Body fossils preserve actual remains like bones, teeth, and shells. Mold fossils form when an organism decomposes but leaves an impression in the rock, like a cookie cutter shape. Cast fossils form when a mold gets filled with minerals, creating a 3D copy. Trace fossils preserve evidence of animal activity — footprints, burrows, and nests. Even ancient poop (called coprolites) can become fossils and tell scientists what animals ate!

What Fossils Tell Us

Fossils are the primary evidence scientists use to understand life on ancient Earth. By studying fossil shapes, sizes, and locations, paleontologists can determine what an organism looked like, what it ate, how it moved, and what environment it lived in. A fish fossil found on a mountain tells us that area was once underwater. Tropical plant fossils found in Antarctica tell us the continent was once warm. Fossils are like puzzle pieces that help us reconstruct Earth's incredible history.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify burial depth as an external component and pressure and fossilization as internal components that change in response to how deeply the remains are buried.

E **Establish Relationships**
Students determine that burial depth positively affects pressure, and pressure positively affects fossilization — deeper burial creates more pressure which drives more complete fossilization.

V **Visualize & Model**
Students build a model showing how increasing burial depth leads to greater pressure and more complete fossilization of remains.

E **Evaluate Outcomes**
Students run shallow, medium, and deep burial scenarios to observe how changing burial depth affects pressure and fossilization across the system.

R **Revise & Extend**
Students add mineral content or time to explore how additional factors affect fossil quality — more minerals and more time produce better-preserved fossils.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"What if I told you that this rock is actually a BONE from an animal that lived 100 million years ago? How is that possible?"*

Watch For

Pass around fossil specimens or high-quality images. Let students examine them closely and share observations.

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to solve the mystery of how ancient bones turn into rock fossils that last FOREVER!"*

Watch For

Read objectives together. Introduce vocabulary — show real sediment (sand, mud) and explain what a paleontologist does.

Slide 3: The Big Question

3 min

Say: *"How does something that was once alive turn into solid rock? That is our mystery to solve!"*

Watch For

Have students turn and talk: What do you THINK happens to bones underground over millions of years? Share ideas with the class.

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model of how fossils form — step by step, layer by layer!"*

Watch For

Preview each LEVER step. Show a simple animation of sediment layers building up over time.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What parts of the fossil formation system can we identify? What starts the process?"*

Watch For

Guide sorting of three components. Explain burial depth is external because natural events like floods and volcanic eruptions cause burial.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When remains are buried deeper, does pressure go up or down? And how does that affect fossilization?"*

Watch For

Help students draw positive arrows from burial depth to pressure, and from pressure to fossilization.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Time to test! What happens when bones are barely buried versus buried super deep? Let us find out!"*

Watch For

Have students predict first, then run each scenario. Compare shallow vs. medium vs. deep burial results.

Slide 8: What Did We Discover?

7 min

Say: *"Now we know the secret — fossils are made by TIME, PRESSURE, and MINERALS working together underground!"*

Watch For

Summarize discoveries. Stack books on a student's hand to demonstrate how pressure increases with depth. Connect to the model.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Burial Depth
- **INTERNAL:** Pressure, Fossilization

Relationships

- **Burial Depth to Pressure:** POSITIVE (+) — The deeper the remains are buried, the more layers of sediment sit on top of them. Each layer adds weight, so deeper burial means much more pressure squeezing down on the remains. Think of stacking heavy books — the bottom book feels the most pressure.
- **Pressure to Fossilization:** POSITIVE (+) — Greater pressure helps compact the sediment into rock and pushes mineral-rich water through the remains. This mineral replacement process works better under high pressure, producing more complete and well-preserved fossils over millions of years.

Key Discoveries

- Fossils form when remains are buried deep under layers of sediment over millions of years
- More burial depth creates more pressure, which helps the fossilization process happen more completely
- Minerals in groundwater slowly replace the original bone or shell material, turning it into rock
- Fossils tell us what animals looked like, what they ate, and what their environment was like long ago

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Fossils are actual bones"	Fossils are NOT the original bones — they are rock copies. The original bone material was slowly replaced by minerals over millions of years. A fossil has the exact shape of the original bone but is made entirely of rock minerals. It is like making a copy of a key — the copy works the same but is made of different material.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Dinosaurs are the only animals that become fossils"	All kinds of organisms can become fossils — fish, insects, plants, shells, even bacteria! In fact, the most common fossils are sea creatures like trilobites and ammonites because ocean sediment buries remains quickly. Dinosaur fossils are actually quite rare compared to marine fossils.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Fossils form quickly, like in a few years"	Fossilization is an incredibly slow process that takes millions of years. The mineral replacement happens molecule by molecule as groundwater slowly seeps through the buried remains. There are no shortcuts — time and pressure are both essential for creating a true fossil.